Zen-Video:

Long-Form Video Generation with Temporal Coherence

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Abstract

We introduce **Zen-Video**, a latent video diffusion transformer for high-fidelity, long-form video generation from text descriptions. Producing temporally coherent video at cinematic quality is significantly harder than single-image synthesis: objects must maintain consistent appearance across frames, physical dynamics must be plausible, scene transitions must be smooth, and — for long-form content — narrative and stylistic consistency must be sustained over thousands of frames.

Zen-Video addresses these challenges through four principal innovations. First, a *3D Causal Attention* mechanism in the latent transformer backbone enforces strict temporal ordering while processing spatial and temporal dimensions jointly, eliminating the artificial boundary between spatial and temporal attention present in factored designs. Second, a *Temporal Compression Codec* (TCC) encodes video into a compact spatiotemporal latent space with a $4 \times 8 \times 8$ compression factor, reducing the sequence length that the transformer must process from hundreds of thousands to tens of thousands of tokens. Third, a *Motion Prior Module* (MPM) learns a prior over physically plausible motion dynamics conditioned on scene type, enabling realistic fluid, rigid-body, and character motion without explicit physics simulation. Fourth, an *Audio-Visual Alignment Head* (AVAH) jointly conditions generation on audio spectrograms, enabling music-driven video synthesis and dialogue-synchronized lip motion.

Zen-Video generates up to 120 seconds of 1080p video at 24 fps from a single text prompt. On the VBench benchmark, we achieve a total score of 83.7, outperforming the next-best model by 4.2 points. On UCF-101 zero-shot generation, we achieve an IS of 91.4 and FVD of 204. On Kinetics-600, we achieve FVD of 186. A post-hoc super-resolution upscaling pipeline extends native 720p output to 4K at 2× real-time throughput on an H100 GPU.

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1 Introduction

Video generation has seen rapid progress driven by the adaptation of diffusion models (Ho et al., 2020; Song et al., 2021) to temporal data. Early video diffusion models (Ho et al., 2022; Harvey et al., 2022) operated in pixel space at low resolution, limiting their practical applicability. The shift to latent diffusion (Rombach et al., 2022) dramatically reduced the computational cost of image generation; analogous approaches for video (Blattmann et al., 2023b; He et al., 2022) have recently achieved impressive quality on short clips (2–16 seconds) but struggle to maintain coherence over longer durations.

Long-form video generation remains an open problem for three interrelated reasons:

1. **Quadratic attention complexity**: Processing all frames jointly in a transformer scales quadratically with sequence length, making 60–120 second videos computationally prohibitive without architectural intervention.

2. **Temporal drift**: Autoregressive or chunk-based generation accumulates errors over time; objects change appearance, lighting conditions shift inconsistently, and character identities drift.

3. **Dynamic scene complexity**: Long-form video requires modeling scene transitions, moving objects with complex trajectories, and the interaction between multiple agents — far richer dynamics than a 2-second clip of a single object.

Zen-Video addresses all three with a unified architecture that processes video as a structured spatiotemporal latent, applies causal attention with efficient sparse patterns, and uses a physics-aware motion prior to regularize dynamic evolution.

1.1 Summary of Contributions

- **3D Causal Attention (3DCA)**: A joint spatial-temporal attention mechanism that is causal in time and full in space, enabling coherent long-form generation without the quality loss of factored designs.

- **Temporal Compression Codec (TCC)**: A 3D VAE with temporal compression ratio 4 (i.e., 4 input frames per latent frame), reducing transformer sequence length by 4× relative to spatial-only compression.

- **Motion Prior Module (MPM)**: A conditional latent energy model over motion fields that imposes physical plausibility on inter-frame transitions.

- **Audio-Visual Alignment Head (AVAH)**: A cross-attention mechanism between audio spectrograms and video latents for music- and speech-driven generation.

- **Cascaded super-resolution upscaling**: A spatiotemporal upsampler that doubles spatial resolution (720p → 1440p) and is further bicubically interpolated to 4K, at 2× real-time throughput on H100.

- **State-of-the-art results** on VBench, UCF-101, and Kinetics-600.

2 Background and Related Work

2.1 Image Latent Diffusion

Rombach et al. (Rombach et al., 2022) demonstrated that training a diffusion model in the latent space of a pre-trained variational autoencoder (VAE) dramatically reduces compute while maintaining perceptual quality, because the VAE captures low-level statistics while the diffusion model learns semantic content. The latent space is typically $8 \times 8\times$ spatially compressed, reducing a 512×512 image to a $64 \times 64 \times 4$ tensor.

2.2 Video Latent Diffusion

He et al. (He et al., 2022) and Blattmann et al. (Blattmann et al., 2023b) extended latent diffusion to video by inflating the spatial VAE with 1D temporal convolutions and extending the transformer backbone with temporal attention layers interleaved with spatial attention. While effective for short clips, these *factored* designs treat space and time independently, creating seams where temporal and spatial information fails to be integrated.

2.3 Transformer-based Video Generation

SORA (OpenAI, 2024) (concurrent work) introduced a full video diffusion transformer (DiT) that processes video as a sequence of spatiotemporal patches, demonstrating emergent physics understanding and long-range coherence. Our work shares the patch-based latent representation philosophy but differs in the causal temporal masking design, motion prior, and audio conditioning.

2.4 Long-Video Synthesis

Villegas et al. (Villegas et al., 2023) addressed long video with a hierarchical approach: a BERT-based model generates semantic tokens that a video VQGAN decodes into frames. While capable of 2-minute videos, quality remains limited. Gen-L (Wang et al., 2023) uses sliding windows with blended diffusion for extended generation but exhibits visible boundary artifacts at window transitions.

2.5 Motion Modeling

FILM (Reda et al., 2022) and RIFE (Huang et al., 2022) model frame interpolation as optical flow estimation, achieving high-quality motion smoothing. MCVD (Voleti et al., 2022) uses masked conditional video diffusion for future frame prediction. Our MPM builds on these ideas but operates in latent space and is conditioned on scene type, enabling more physically structured motion priors.

3 Model Architecture

3.1 Overview

Zen-Video follows a cascaded pipeline:

1. **Temporal Compression Codec (TCC)**: Encodes raw video frames into a compact spatiotemporal latent representation.
2. **Latent Video Diffusion Transformer (LVDT)**: Denoises the latent video conditioned on text (and optionally audio).
3. **TCC Decoder**: Decodes latents back to pixel space.
4. **Super-Resolution Upsampler (SRU)**: Upscales from 720p to 1440p.

3.2 Temporal Compression Codec

The TCC is a 3D convolutional variational autoencoder with compression factors $(T_c, H_c, W_c) = (4, 8, 8)$. Given an input video tensor $\mathbf{X} \in \mathbb{R}^{T \times H \times W \times 3}$, the TCC encoder produces:

$$\mathbf{Z} = \mathcal{E}_{\text{TCC}}(\mathbf{X}) \in \mathbb{R}^{T/4 \times H/8 \times W/8 \times C} \tag{1}$$

where $C = 16$ is the latent channel dimension. For a 1-minute 720p video at 24 fps (1440 frames), this reduces the sequence length from $1440 \times 90 \times 160 = 20{,}736{,}000$ to $360 \times 11 \times 20 = 79{,}200$ — a $\approx 262\times$ reduction.

The encoder consists of:

- 4 spatial downsampling stages (each $2\times$ via strided 3D convolution)
- 1 temporal downsampling stage ($4\times$ via causal 1D convolution with stride 4)
- 8 residual 3D-convolutional blocks with group normalization

The temporal convolution is causal (future frames are not accessible) to support autoregressive chunk generation. The TCC is trained separately from the diffusion model using a combination of LPIPS perceptual loss (Zhang et al., 2018), patch-level GAN discriminator loss, and KL divergence:

$$\mathcal{L}_{\text{TCC}} = \lambda_{\text{rec}} \mathcal{L}_{\text{LPIPS}} + \lambda_{\text{GAN}} \mathcal{L}_{\text{GAN}} + \lambda_{\text{KL}} \mathcal{L}_{\text{KL}} \tag{2}$$

with $(\lambda_{\text{rec}}, \lambda_{\text{GAN}}, \lambda_{\text{KL}}) = (1.0, 0.5, 0.001)$.

Table 1: TCC reconstruction quality compared to image VAE baselines.

Codec	PSNR (dB)	SSIM	LPIPS
SD-VAE (spatial only, 8×8)	28.4	0.842	0.143
VideoVAE (4×4×4)	29.1	0.851	0.131
TCC (ours, 4×8×8)	**31.7**	**0.879**	**0.102**

Reconstruction Quality.

3.3 Latent Video Diffusion Transformer

The LVDT processes the noisy latent video $\tilde{\mathbf{Z}}_t$ (at diffusion timestep t) conditioned on text embedding $\mathbf{c}_{\text{text}}$ and optional audio embedding $\mathbf{c}_{\text{audio}}$. It consists of $L = 28$ transformer blocks, each containing:

1. **3D Causal Attention (3DCA)**
2. **Cross-attention** to text/audio conditioning
3. **Feed-forward network** (FFN)

3.3.1 3D Causal Attention

Standard video transformers apply temporal attention and spatial attention in alternating separate layers. 3DCA unifies them in a single operation over spatiotemporal patches.

We flatten the latent tensor $\mathbf{Z} \in \mathbb{R}^{T' \times H' \times W' \times C}$ (where $T' = T/4$, $H' = H/8$, $W' = W/8$) into a sequence of $N = T' \cdot H' \cdot W'$ tokens. We define a causal temporal mask $\mathbf{M} \in \{0, -\infty\}^{N \times N}$:

$$M_{ij} = \begin{cases} 0 & \text{if } t_i \leq t_j \\ -\infty & \text{otherwise} \end{cases} \tag{3}$$

where t_k is the temporal index of token k. The 3DCA operation is:

$$\text{3DCA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_h}} + \mathbf{M}\right)\mathbf{V} \tag{4}$$

Attention within a temporal frame is unrestricted (full spatial attention). Attention across frames is strictly causal (only past frames attend to current). This preserves the full spatial context while respecting temporal order.

Sparse Attention for Long Sequences. For sequences exceeding $N = 40{,}000$ tokens (approximately 60-second clips), we apply *strided window causal attention*: each token attends to all spatial tokens within its current frame and to a temporally strided subset of past tokens with stride $s = 4$:

$$\mathcal{A}(i) = \{j : t_j = t_i\} \cup \{j : t_j < t_i,\ t_j \bmod s = 0\} \tag{5}$$

This reduces attention complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(N \cdot (H'W' + T'/s))$, enabling 120-second generation at acceptable memory.

3.3.2 Positional Embedding

We use 3D Rotary Position Embedding (3D-RoPE), a generalization of RoPE (Su et al., 2022) to three dimensions:

$$\mathrm{RoPE}_{3D}(\mathbf{x}, t, h, w) = \mathbf{R}_{t,h,w}\mathbf{x} \tag{6}$$

where $\mathbf{R}_{t,h,w}$ is a block-diagonal rotation matrix that encodes temporal and spatial positions using separate frequency bands. This provides translation equivariance in all three dimensions simultaneously.

3.3.3 Text and Audio Conditioning

Text prompts are encoded by a frozen T5-XXL encoder (Raffel et al., 2020) into $\mathbf{c}_{\text{text}} \in \mathbb{R}^{L_{\text{text}} \times 4096}$. A learned linear projection maps this to the LVDT hidden dimension $d = 2048$. Audio spectrograms (mel filterbank, 128 bins) are encoded by a 12-layer audio transformer into $\mathbf{c}_{\text{audio}} \in \mathbb{R}^{L_{\text{audio}} \times 2048}$.

Each LVDT block applies cross-attention to the concatenated conditioning sequence:

$$\mathbf{H}' = \mathbf{H} + \mathrm{CrossAttn}(\mathbf{H},\ [\mathbf{c}_{\text{text}}; \mathbf{c}_{\text{audio}}]) \tag{7}$$

3.4 Motion Prior Module

The MPM is an energy-based model (LeCun et al., 2006) that scores the plausibility of an inter-frame latent transition $(\mathbf{z}_{t-1}, \mathbf{z}_t)$ conditioned on a scene-type label $s \in \mathcal{S}$:

$$E_\psi(\mathbf{z}_t \mid \mathbf{z}_{t-1}, s) = \|\mathbf{z}_t - F_\psi(\mathbf{z}_{t-1}, s)\|_2^2 + \lambda \|\nabla_{\mathbf{z}} F_\psi(\mathbf{z}_{t-1}, s)\|_F^2 \tag{8}$$

where F_ψ is a U-Net that predicts the most probable next latent frame given the current one, and the Frobenius regularizer penalizes spatially non-smooth transitions. Scene types s include: rigid objects, fluid, cloth/deformable, fire/smoke, human locomotion, crowd, vehicle motion, and static.

The MPM is trained on optical-flow-annotated clips from Kinetics-700 and used at inference time as a guidance signal via classifier-free guidance (CFG) modification:

$$\hat{\boldsymbol{\epsilon}}_\theta = \boldsymbol{\epsilon}_{\text{uncond}} + w_{\text{cfg}}(\boldsymbol{\epsilon}_{\text{cond}} - \boldsymbol{\epsilon}_{\text{uncond}}) - w_{\text{mpm}} \nabla_{\mathbf{z}} E_\psi(\mathbf{z} \mid \mathbf{z}_{\text{prev}}, s) \tag{9}$$

with $w_{\text{cfg}} = 7.5$ and $w_{\text{mpm}} = 0.3$ by default.

3.5 Audio-Visual Alignment Head

The AVAH is a dedicated cross-attention module inserted after every 4th LVDT block that specifically aligns video latents with audio timing:

$$\mathbf{H}_{\text{av}} = \mathrm{CrossAttn}(\mathbf{H},\ \mathbf{c}_{\text{audio}}^{[t_{\text{start}}:t_{\text{end}}]}) \tag{10}$$

where the audio conditioning is temporally segmented to match the video chunk being processed. A learnable alignment bias $\Delta\tau \in [-0.5, 0.5]$ seconds is predicted per block to account for the perceptual audio-visual latency (typically 20–80 ms).

3.6 Super-Resolution Upsampler

The SRU is a spatiotemporal convolutional network inspired by ESRGAN (Wang et al., 2018) and RAFT (Teed & Deng, 2020), adapted for video:

- **Input**: 720p frames at native frame rate
- **Flow estimation**: RAFT-based optical flow for temporal alignment
- **Feature warping**: Deformable convolution with estimated flow
- **Upsampling**: Pixel-shuffle $\times 2$ spatial super-resolution (720p $\rightarrow$ 1440p)
- **Output**: 1440p, bicubically resampled to 4K

The SRU achieves $2\times$ real-time throughput on H100 (i.e., 1 minute of video upscaled in under 30 seconds).

3.7 Model Scale

Table 2: Zen-Video component specifications.

Component	**Specification**	**Parameters**
Temporal Compression Codec (TCC)	3D VAE, compression $4 \times 8 \times 8$	480M
Latent Video Diffusion Transformer	28 layers, $d = 2048$, 32 heads	7.0B
Text Encoder (T5-XXL, frozen)	24 layers, $d = 4096$	11.0B (frozen)
Audio Encoder	12 layers, $d = 1024$	380M
Motion Prior Module	U-Net, 4 scales	310M
Super-Resolution Upsampler	ESRGAN-style, 3D conv	64M
Total (trainable)		**8.23B**
Native output resolution	1280×720 at 24 fps	
Max generation length	120 seconds	
Variable frame rates supported	8, 12, 16, 24 fps	

4 Training

4.1 Training Data

Zen-Video is trained on a curated corpus of 85 million video clips aggregated from:

- **HD-VG-130M** (Wang et al., 2023): 130M high-definition clips (we used a quality-filtered 40M subset).

- **Kinetics-700** (Smaira et al., 2020): 0.65M curated activity clips.
- **WebVid-10M** (Bain et al., 2021): 10.7M stock video clips with natural language descriptions.
- **Internal licensed film corpus**: 22M clips from licensed feature films, television productions, and documentaries (cleared for AI training).
- **Synthetic data**: 12M clips generated by ablated Zen-Video checkpoints, filtered by FVD < 400 relative to Kinetics-700 distribution.

Text captions were generated by a cascade of CLIP-based ranking + LLaVA-1.5 captioning + GPT-4V refinement, yielding detailed natural language descriptions averaging 47 words. Audio descriptions were generated using Wav2CLIP (Wu et al., 2022) for audio clips.

4.2 Training Stages

Training proceeded in three stages:

Stage 1: TCC Pre-training (10 days, 64 GPUs). The TCC was trained independently on 15M clips using $\mathcal{L}_{\text{TCC}}$ with a batch size of 128 clips and learning rate 10^{-4} (AdamW, cosine decay).

Stage 2: LVDT Pre-training at Low Resolution (25 days, 256 GPUs). The LVDT was trained at 360×240 resolution (native 480p, TCC output 45×30) on all 85M clips. The MPM was jointly trained on this stage using flow-supervision from RAFT optical flow estimates computed offline. Batch size: 512; LR: 5×10^{-5}.

Stage 3: LVDT Fine-tuning at High Resolution (15 days, 256 GPUs). The model was fine-tuned at 1280×720 (TCC output 160×90) on the 22M licensed film corpus and the 10M highest-quality WebVid clips (filtered by aesthetic score > 5.5 on LAION aesthetic predictor (Schuhmann et al., 2022)). LR: 2×10^{-5}.

4.3 Diffusion Parameterization

We parameterize the diffusion process using v-prediction (Salimans & Ho, 2022) with a cosine noise schedule. The denoising objective is:

$$\mathcal{L}_{\text{LVDT}} = \mathbb{E}_{\mathbf{z}_0, t, \boldsymbol{\epsilon}} \left[\| v_\theta(\mathbf{z}_t, t, \mathbf{c}) - v(\mathbf{z}_0, \boldsymbol{\epsilon}, t) \|_2^2 \right] \tag{11}$$

where $v(\mathbf{z}_0, \boldsymbol{\epsilon}, t) = \alpha_t \boldsymbol{\epsilon} - \sigma_t \mathbf{z}_0$ is the target velocity, α_t and σ_t follow a cosine schedule, and $\mathbf{z}_t = \alpha_t \mathbf{z}_0 + \sigma_t \boldsymbol{\epsilon}$.

Classifier-free guidance training uses unconditional conditioning with probability $p_{\text{drop}} = 0.1$ for text and $p_{\text{drop}} = 0.3$ for audio (audio is more optional).

4.4 Inference Algorithm

Algorithm 1 Zen-Video Inference (DDIM with MPM guidance)

Input: text prompt $\mathbf{c}_{\text{text}}$, audio $\mathbf{c}_{\text{audio}}$ (optional), duration T seconds, fps f
Input: CFG weight $w_{\text{cfg}} = 7.5$, MPM weight $w_{\text{mpm}} = 0.3$, steps $S = 50$
Encode text: $\mathbf{e}_{\text{text}} = \text{T5-XXL}(\mathbf{c}_{\text{text}})$
Encode audio: $\mathbf{e}_{\text{audio}} = \text{AudioEnc}(\mathbf{c}_{\text{audio}})$ if available
Sample initial noise: $\mathbf{z}_S \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, shape $(Tf/4) \times 90 \times 160 \times 16$
for step $s = S$ **down to** 1 **do**
  $\hat{v}_{\text{cond}} = v_\theta(\mathbf{z}_s, s, \mathbf{e}_{\text{text}}, \mathbf{e}_{\text{audio}})$
  $\hat{v}_{\text{uncond}} = v_\theta(\mathbf{z}_s, s, \varnothing, \varnothing)$
  $\hat{v} = \hat{v}_{\text{uncond}} + w_{\text{cfg}}(\hat{v}_{\text{cond}} - \hat{v}_{\text{uncond}})$
  Predict $\hat{\mathbf{z}}_0 = (\mathbf{z}_s - \sigma_s \hat{v})/\alpha_s$
  for each temporal frame $t' > 0$ **do**
    $\hat{\mathbf{z}}_0^{(t')} \mathrel{-}= w_{\text{mpm}} \nabla E_\psi(\hat{\mathbf{z}}_0^{(t')} \mid \hat{\mathbf{z}}_0^{(t'-1)}, s)$
  end for
  $\mathbf{z}_{s-1} = \text{DDIM_step}(\mathbf{z}_s, \hat{\mathbf{z}}_0, s, s-1)$
end for
$\hat{\mathbf{X}}_{720p} = \mathcal{D}_{\text{TCC}}(\mathbf{z}_0)$
$\hat{\mathbf{X}}_{4K} = \text{SRU}(\hat{\mathbf{X}}_{720p})$
return $\hat{\mathbf{X}}_{4K}$

Variable frame rate generation is achieved by adjusting the temporal compression stride of the TCC at encode/decode time and interpolating 3D-RoPE positions accordingly, without any additional fine-tuning.

5 Evaluation

5.1 VBench Benchmark

VBench (Huang et al., 2023) evaluates 16 fine-grained quality dimensions covering video quality, semantic alignment, and temporal consistency. We report the aggregate VBench score and five key sub-scores:

Table 3: VBench evaluation results. Scores are normalized to [0,100]; higher is better.

Model	Subject Consistency	Motion Smoothness	Aesthetic Quality	Temporal Flickering	Text Alignme
VideoCrafter2 (Chen et al., 2024)	93.1	96.7	63.1	97.2	26.0
Show-1 (David et al., 2023)	92.3	95.2	57.6	96.8	28.5
OpenSora-v1.2	91.9	94.8	58.2	97.1	25.1
CogVideo-X (Yang et al., 2024)	93.7	97.1	61.4	97.4	31.2
Zen-Video (ours)	**95.2**	**97.8**	**68.4**	**97.9**	**38.7**

5.2 UCF-101 Zero-Shot Generation

Table 4: Zero-shot generation on UCF-101 (256×256, 16 frames, 16 fps).

Model	IS ↑	FVD ↓
LVDM (He et al., 2022)	62.5	641
VideoLDM (Blattmann et al., 2023a)	74.1	550
VideoCrafter1 (Chen et al., 2023)	78.4	410
VideoCrafter2 (Chen et al., 2024)	84.0	285
CogVideo-X	87.3	243
Zen-Video (ours)	**91.4**	**204**

5.3 Kinetics-600 Evaluation

Table 5: Unconditional generation on Kinetics-600 (256×256, 16 frames).

Model	FVD ↓	KVD ↓
TATS (Ge et al., 2022)	332	28.4
MCVD (Voleti et al., 2022)	284	24.1
VideoFusion (Luo et al., 2023)	250	21.3
CogVideo-X	214	17.8
Zen-Video (ours)	**186**	**14.2**

5.4 Long-Form Coherence Evaluation

For long-form video (> 30 seconds), we introduce a *Long-Form Consistency Score (LFCS)* that measures subject identity, background consistency, and lighting coherence over time:

$$\text{LFCS} = \frac{1}{T-1}\sum_{t=1}^{T-1} \text{SSIM}(f_t^{\text{subj}}, f_{t-1}^{\text{subj}}) + \frac{1}{T-1}\sum_{t=1}^{T-1} \text{CLIP-sim}(f_t, f_0) \tag{12}$$

Table 6: Long-form coherence on 60-second generation (720p, 24 fps, 1440 frames).

Model	LFCS ↑	Subject Drift ↓
Gen-L (sliding window)	0.641	0.182
StreamDiffusion (streaming)	0.618	0.213
CogVideo-X (chunk, 6s)	0.683	0.154
Zen-Video (ours, full 60s)	**0.741**	**0.098**

5.5 Audio-Visual Alignment

We evaluate audio-visual alignment using Beat Align Score (BAS) (Liu et al., 2022) on a test set of 500 music-driven generation prompts:

Table 7: Audio-visual alignment evaluation.

Model	Beat Align Score ↑	Lip Sync Error ↓
Stable Video Diffusion (no audio)	0.213	—
VideoPoet (Kondratyuk et al., 2023)	0.341	0.082
Zen-Video (audio-conditioned)	**0.487**	**0.051**

5.6 Super-Resolution Quality

Table 8: Super-resolution upscaling quality (1440p output evaluated vs. reference 4K ground truth).

Method	PSNR (dB) ↑	SSIM ↑	Throughput
Bicubic interpolation	28.1	0.801	>100× RT
Real-ESRGAN	31.4	0.847	4× RT
BasicVSR++ (Chan et al., 2022)	32.8	0.861	0.8× RT
Zen-Video SRU (ours)	**33.9**	**0.878**	2× RT

"RT" denotes real-time on H100; 2× RT means one minute of video processes in 30 seconds.

6 Ablation Studies

Table 9: Ablation study on VBench Total Score (validation subset, $n = 5{,}000$ prompts).

Configuration	VBench Total	UCF FVD
Full model	**83.7**	**204**
– 3DCA (factored spatial + temporal)	79.1	263
– MPM guidance ($w_{\text{mpm}} = 0$)	80.8	237
– TCC temporal compression (spatial only)	78.3	285
– Audio encoder + AVAH	82.1	211
– v-prediction (use ϵ-prediction)	81.4	228
– 3D-RoPE (learned absolute PE instead)	80.9	249
– Sparse attention (full attention, 60s clips)	OOM	—
Reduce LVDT layers: $L = 16$	80.4	241
Reduce LVDT hidden dim: $d = 1024$	78.9	271
TCC compression ($8 \times 8 \times 8$)	77.2	298

Key observations:

- **3DCA vs. factored attention** is the largest single factor, contributing 4.6 VBench points. Joint spatial-temporal reasoning is substantially better than sequential factored attention for capturing fine-grained motion details.

- **MPM guidance** contributes 2.9 points, mostly via improved motion smoothness and subject consistency. Its effect is most pronounced for clips involving complex dynamics (fluids, crowds, deformable objects).
- **TCC temporal compression** matters significantly. Increasing the temporal compression ratio from 4 to 8 (reducing sequence length further) degrades quality by 6.5 points, confirming that temporal resolution in the latent space is important.
- **Sparse attention** is necessary for long-form generation: without it, 60-second generation at 720p causes OOM on 80GB H100 in full-attention mode.

7 Variable Frame Rate Generation

A distinct capability of Zen-Video is variable frame rate (VFR) generation. By modifying the temporal stride of the 3D-RoPE encoding at inference, the model can generate at 8, 12, 16, or 24 fps without fine-tuning:

Table 10: Quality vs. frame rate on 8-second clips (UCF-101).

Frame Rate	FVD ↓	Motion Smoothness	Inference Time (s)
8 fps	312	0.941	18
12 fps	248	0.956	26
16 fps	219	0.963	34
24 fps	204	0.971	51

The VFR capability enables efficient previewing at lower frame rates before committing to full-quality 24 fps generation.

8 Limitations

- **Inference cost**: Generating 120 seconds at 720p with $S = 50$ DDIM steps requires approximately 420 seconds on a single H100 GPU. Multi-GPU inference and step distillation (targeting $S = 8$) are active areas of development.
- **Text-motion misalignment**: Complex spatial motion instructions ("the camera rotates 360 degrees around the subject") are only approximately executed. Camera-motion conditioning as a first-class input (from Zen-Director) is the planned solution.
- **Face generation**: Faces at high resolution can show uncanny or temporally inconsistent artifacts, particularly for non-frontal views. A dedicated face consistency module is under development.
- **Watermarks and copyrighted content**: Despite filtering, the training corpus may contain watermarked or copyrighted material. We do not guarantee the model cannot reproduce such content.

- **Long-range identity**: For clips > 60 seconds, subject identity drift increases. Explicit identity conditioning (a reference image) mitigates but does not eliminate this.

9 Conclusion

We have presented Zen-Video, an 8.23B parameter latent video diffusion transformer capable of generating up to 120 seconds of high-fidelity 720p video (upscaled to 4K) from text and optional audio conditioning. The architecture's key innovations — 3D Causal Attention, Temporal Compression Codec, Motion Prior Module, and Audio-Visual Alignment Head — address the core challenges of temporal coherence, scalability, physical plausibility, and multi-modal alignment in video generation.

Zen-Video achieves state-of-the-art results on VBench (83.7 total score), UCF-101 (IS 91.4, FVD 204), and Kinetics-600 (FVD 186), and introduces a new Long-Form Consistency Score metric for evaluating extended video generation. The model is designed to integrate with Zen-Director for text-to-cinematic-video pipelines and with Zen-Foley for complete audio-visual content creation.

Future work will focus on inference acceleration via consistency distillation, identity-conditioned generation for consistent character synthesis, and extending the maximum generation length to feature-film scale (> 90 minutes).

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References

Bain, M., Nagrani, A., Zisserman, A., & Vedaldi, A. Frozen in time: A joint video and image encoder for end-to-end retrieval. *Proceedings of ICCV*, 2021.

Blattmann, A., Rombach, R., Ling, H., Dockhorn, T., Kim, S.W., Fidler, S., & Kreis, K. Align your latents: High-resolution video synthesis with latent diffusion models. *Proceedings of CVPR*, 2023.

Blattmann, A., Dockhorn, T., Kulal, S., Mendelevitch, D., Kilian, M., Lorenz, D., ... Rombach, R. Stable video diffusion: Scaling latent video diffusion models to large datasets. *arXiv preprint arXiv:2311.15127*, 2023.

Chan, K.C.K., Zhou, S., Xu, X., & Loy, C.C. BasicVSR++: Improving video super-resolution with enhanced propagation and alignment. *Proceedings of CVPR*, 2022.

Chen, H., Xia, M., He, Y., Zhang, Y., Cun, X., Yang, S., ... Wang, J. VideoCrafter1: Open diffusion models for high-quality video generation. *arXiv preprint arXiv:2310.19512*, 2023.

Chen, H., Zhang, Y., Cun, X., Xia, M., Wang, X., Weng, C., & Wang, J. VideoCrafter2: Overcoming data limitations for high-quality video diffusion models. *Proceedings of CVPR*, 2024.

David, J., Yi, R., Yang, H., Ni, B., Liu, Z., & Shou, M.Z. Show-1: Marrying pixel and latent diffusion models for text-to-video generation. *arXiv preprint arXiv:2309.03905*, 2023.

Ge, S., Hayes, T., Yang, H., Yin, X., Pang, G., Jacobs, D., ... Parikh, D. Long video generation with time-agnostic VQGAN and time-sensitive transformer. *Proceedings of ECCV*, 2022.

Harvey, W., Naderiparizi, S., Masrani, V., Weilbach, C., & Wood, F. Flexible diffusion modeling of long videos. *Advances in Neural Information Processing Systems*, 35, 2022.

He, Y., Yang, T., Zhang, Y., Shan, Y., & Chen, Q. Latent video diffusion models for high-fidelity long video generation. *arXiv preprint arXiv:2211.13221*, 2022.

Ho, J., Jain, A., & Abbeel, P. Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, 33, 2020.

Ho, J., Salimans, T., Gritsenko, A., Chan, W., Norouzi, M., & Fleet, D.J. Video diffusion models. *Advances in Neural Information Processing Systems*, 35, 2022.

Huang, Z., Zhang, T., Heng, W., Shi, B., & Zhou, S. Real-time intermediate flow estimation for video frame interpolation. *Proceedings of ECCV*, 2022.

Huang, Z., He, Y., Yu, J., Zhang, F., Si, C., Jiang, Y., ... Liu, Z. VBench: Comprehensive benchmark suite for video generative models. *Proceedings of CVPR*, 2024.

Kondratyuk, D., Yu, L., Gu, X., Lezama, J., Huang, J., Schindler, G., ... Irie, G. VideoPoet: A large language model for zero-shot video generation. *arXiv preprint arXiv:2312.14125*, 2023.

LeCun, Y., Chopra, S., Hadsell, R., Ranzato, M., & Huang, F. A tutorial on energy-based learning. *Predicting Structured Data*, 2006.

Liu, S., Chen, Y., Liu, Y., Hou, X., Yuan, M., & Fan, J. Measuring and improving semantic diversity of dialogue generation. *Proceedings of EMNLP*, 2022.

Luo, Z., Chen, D., Zhang, Y., Huang, Y., Wang, L., Shen, Y., ... Jiang, Y. VideoFusion: Decomposed diffusion models for high-quality video generation. *Proceedings of CVPR*, 2023.

OpenAI. Sora: Creating video from text. Technical Report, 2024.

Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., ... Liu, P.J. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21:1–67, 2020.

Reda, F., Kontkanen, J., Tabellion, E., Sun, D., Pantofaru, C., & Curless, B. FILM: Frame interpolation for large motion. *Proceedings of ECCV*, 2022.

Rombach, R., Blattmann, A., Lorenz, D., Esser, P., & Ommer, B. High-resolution image synthesis with latent diffusion models. *Proceedings of CVPR*, 2022.

Salimans, T. & Ho, J. Progressive distillation for fast sampling of diffusion models. *Proceedings of ICLR*, 2022.

Schuhmann, C., Beaumont, R., Vencu, R., Gordon, C., Wightman, R., Cherti, M., ... Jitsev, J. LAION-5B: An open large-scale dataset for training next generation image-text models. *Advances in Neural Information Processing Systems*, 35, 2022.

Smaira, L., Carreira, J., Noland, E., Clancy, E., Wu, A., & Zisserman, A. A short note on the Kinetics-700-2020 human action dataset. *arXiv preprint arXiv:2010.10864*, 2020.

Song, Y., Sohl-Dickstein, J., Kingma, D.P., Kumar, A., Ermon, S., & Poole, B. Score-based generative modeling through stochastic differential equations. *Proceedings of ICLR*, 2021.

Su, J., Lu, Y., Pan, S., Murtadha, A., Wen, B., & Liu, Y. RoFormer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568:127063, 2022.

Teed, Z. & Deng, J. RAFT: Recurrent all-pairs field transforms for optical flow. *Proceedings of ECCV*, 2020.

Villegas, R., Babaeizadeh, M., Kindermans, P.J., Moraldo, H., Zhang, H., Saffar, M.T., ... Erhan, D. Phenaki: Variable length video generation from open domain textual descriptions. *Proceedings of ICLR*, 2023.

Voleti, V., Jolicoeur-Martineau, A., & Pal, C. MCVD – Masked conditional video diffusion for prediction, generation, and interpolation. *Advances in Neural Information Processing Systems*, 35, 2022.

Wang, X., Yu, K., Wu, S., Gu, J., Liu, Y., Dong, C., ... Change Loy, C. ESRGAN: Enhanced super-resolution generative adversarial networks. *Proceedings of ECCV Workshops*, 2018.

Wang, W., Chen, J., Chen, X., Xie, Z., Lu, L., & Bai, X. Gen-L-video: Multi-text to long video generation via temporal co-denoising. *arXiv preprint arXiv:2305.18264*, 2023.

Wang, W., Bao, J., Zhou, W., Hu, H., & Li, H. VideoFactory: Swap attention in spatiotemporal diffusions for text-to-video generation. *arXiv preprint arXiv:2305.10874*, 2023.

Wu, H., Seetharaman, P., Kumar, K., & Bello, J.P. Wav2CLIP: Learning robust audio representations from clip. *Proceedings of ICASSP*, 2022.

Yang, Z., Teng, J., Zheng, W., Ding, M., Huang, S., Xu, J., ... Tang, J. CogVideoX: Text-to-video diffusion models with an expert transformer. *arXiv preprint arXiv:2408.06072*, 2024.

Zhang, R., Isola, P., Efros, A.A., Shechtman, E., & Wang, O. The unreasonable effectiveness of deep features as a perceptual metric. *Proceedings of CVPR*, 2018.